

Lattice dynamics

8

8.1 Why do we need to consider dynamics?

Up to this point in the book we have paid no attention to the motions of atoms in solids. Without so doing we have achieved a considerable level of understanding of the properties and behaviour of solids. There are, however, a number of issues that we cannot begin to tackle without considering dynamics. These include the whole area of thermodynamics, including phase stability. For example, the stable phase of any material is that for which the **free energy** is a minimum. The free energy G has two parts, due to the enthalpy H (assuming constant pressure) and entropy S :

$$G = H - TS \quad (8.1)$$

For many solids the primary origin of entropy is from **thermal vibrations**. Neglect of dynamics is tantamount to asserting that temperature is zero, and hence that entropy is also zero. It is only by including dynamics that we are able to study issues associated with phase stability under changing temperature, such as the ferroelectric phase transitions we have met in the earlier parts of the book.

Thermal expansion and pyroelectricity are two important issues that depend directly on dynamics through their dependence on temperature. The piezoelectric properties of ferroelectric crystals also have a strong dependence on temperature, which can only be understood through the development of a model for the dynamics. In effect, without an understanding of the dynamics of crystals we are unable to develop quantitative models for crystal behaviour that incorporate temperature as a variable.

Our route into the subject is through a few simple models. The simple models can be generalized for application to real materials (and we do so in this chapter), but in keeping them simple we will be able to deduce the important general principles with minimum of effort.

8.2 The harmonic approximation

The ideas developed in this chapter are based on a simple approximation called the **harmonic approximation**. In principle, if $u_{j,\alpha}$ describes a component of the vector displacement ($\alpha = x, y, z$) of the atom labelled j from its equilibrium

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position, the energy can be expressed as a Taylor expansion:

$$\begin{aligned}
 E = E_0 + \frac{1}{2} \sum_{\substack{j,j' \\ \alpha,\alpha'}} \frac{\partial^2 E}{\partial u_{\alpha,j} \partial u_{\alpha',j'}} u_{\alpha,j} u_{\alpha',j'} + \cdots \\
 + \frac{1}{n!} \sum_{\substack{j^{(1)}, \dots, j^{(n)} \\ \alpha^{(1)}, \dots, \alpha^{(n)}}} \frac{\partial^n E}{\partial u_{\alpha^{(1)},j^{(1)}} \cdots \partial u_{\alpha^{(n)},j^{(n)}}} u_{\alpha^{(1)},j^{(1)}} \cdots u_{\alpha^{(n)},j^{(n)}} \\
 + \cdots
 \end{aligned} \tag{8.2}$$

E_0 is the equilibrium lattice energy (Chapter 5), which we stress again is the energy with all atoms at rest. There is no first-order term in the expansion because by the definition of equilibrium all residual forces are zero. The second-order term is the **harmonic energy**. All higher-order terms are classed together as the **anharmonic energy**. The essence of **harmonic approximation** is to neglect all the anharmonic terms.

The harmonic approximation may appear to be quite crude (brutal even). Its merit in our immediate context is that it gives a mathematical model that has exact solutions. The anharmonic model, even if truncated at the first additional term, cannot be solved exactly (and to truncate after an odd-order term allows the energy to be lowered to a value of $-\infty$). Where the harmonic model has a particular merit is that the anharmonic terms can be introduced as a modification to the harmonic model, provided that the atomic displacements are not large. This condition is often fulfilled. The harmonic model gives many of the features of lattice dynamics, such as the dependence of frequency on wave vector, and the features that are not explained by the harmonic model (such as thermal expansion) can be incorporated into a model that looks like a harmonic model. Thus the harmonic approximation does rather more than simply provide a model that can be solved – it provides the whole framework for the study of lattice dynamics into which improvements can be incorporated with little damage to the framework.

8.3 Lattice vibrations of one-dimensional monatomic crystals

8.3.1 The linear chain model

The simplest illustrative model is a chain of atoms, each of mass m , and separated by the unit cell length a , as shown in Fig. 8.1. Initially we consider that each atom only feels the force of its immediate neighbour, which we will assume to be harmonic. Writing u_n as the displacement of the n -th atom from its equilibrium position, the total energy of a chain of N atoms is

$$E = N E_{\text{bond}} + \frac{1}{2} J \sum_{i=1}^N (u_n - u_{n+1})^2 \tag{8.3}$$

where the force constant J is given by

$$J = \left. \frac{\partial^2 E}{\partial u^2} \right|_{u=0} \tag{8.4}$$

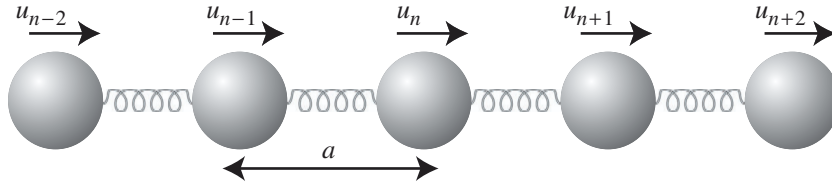


Fig. 8.1 Linear chain model. J is the harmonic force constant (the interactions are represented by springs), and the atomic displacements are represented by u .

The force constant is identical for each pair of atoms in the chain. In forming the differential we note that the distance between two atoms n and $n + 1$ is $r = a + (u_n - u_{n+1})$. Thus the derivatives of E with respect to u are equivalent to the derivatives with respect to r . Since a is the equilibrium unit cell length, the first derivative of E at $r = a$ is zero.

8.3.2 Sound waves – vibrations with long wavelengths

Vibrations with long wavelengths, and hence small wave vectors, will correspond to sound waves. There is a simple relationship between the speed of sound, v , and the isotropic elastic stiffness c :

$$v^2 = c/\rho \quad (8.5)$$

where ρ is the density, which in our one-dimensional model is equal to $\rho = m/a$. The one-dimensional stiffness in the model is easy to compute: a strain ϵ gives a displacement $u_n = \epsilon na$ to atom n , so the strain energy *per unit length* is given as

$$\begin{aligned} E_{\text{strain}} &= \frac{1}{2} \frac{J(u_n - u_{n+1})^2}{a} \\ &= \frac{1}{2} J \epsilon^2 a \equiv \frac{1}{2} c \epsilon^2 \end{aligned} \quad (8.6)$$

The last part of this equation defines the stiffness c :

$$c = Ja \quad (8.7)$$

We therefore have

$$v^2 = Ja^2/m \quad (8.8)$$

The velocity of the sound wave is related to the angular frequency and wave vector through

$$\omega = vk \quad (8.9)$$

Thus we have an equation for the dependence of angular frequency on wave vector:

$$\omega = (a\sqrt{J/m})k \quad (8.10)$$

We have therefore related the vibration frequency to the second derivative of the interatomic potential, and obtained the dependence of the frequency on wave vector in the limit of small k .

8.3.3 Vibrations with shorter wavelengths: general features

We start from the energy of atom n :

$$E_n = \frac{1}{2}J(u_{n+1} - u_n)^2 + \frac{1}{2}J(u_n - u_{n-1})^2 \quad (8.11)$$

The force on the n -th atom (which is given by the differential of the energy) is linked to its acceleration by the classical Newton equation of motion, *force* = *mass* $\times$ *acceleration*:

$$\begin{aligned} F_n &= -\frac{\partial E_n}{\partial u_n} = J(u_{n+1} - u_n) - J(u_n - u_{n-1}) \\ &= -J(2u_n - u_{n+1} - u_{n-1}) = m \frac{\partial^2 u_n}{\partial t^2} \end{aligned} \quad (8.12)$$

The solution of the harmonic equation of motion is a sinusoidal wave, so the motion of the whole system will correspond to a set of travelling waves. Our aim then is to find the set of frequencies of these waves. A wave of a given wave vector k and angular frequency ω_k will cause atom n to be displaced according to the travelling wave equation

$$u_{n,k} = \tilde{u}_k \exp[i(kx_n - \omega t)] \quad (8.13)$$

where k is any wave vector ($= 2\pi/\text{wavelength}$), ω_k is the corresponding angular frequency, $\tilde{u}_k$ is the amplitude, and $x_n = na$ is the position of the n -th atom. From the outset we use the complex exponential form instead of the simpler cosine or sine function, because it makes the manipulation of the differential equations more transparent. We note for future reference that the displacements of the atom's neighbours are given by

$$u_{n\pm 1,k} = u_{n,k} \exp(\pm ika) \quad (8.14)$$

We will find later that there is a discrete set of allowed values of k . We expect the time-dependent motion of the n -th atom to be a linear superposition of each of the travelling waves allowed along the chain; the mathematical representation is

$$u_n(t) = \sum_k \tilde{u}_k \exp[i(kna - \omega t)] \quad (8.15)$$

8.3.4 Vibrations with shorter wavelengths: the special case of $\lambda=2a$

We consider first the special case of a vibration where the wavelength is equal to twice the atom repeat distance, which means that $k = \pi/a$. In this case, neighbouring atoms will move by equal amounts in opposite directions, i.e. $u_n = -u_{n+1}$ (consistent with eqn 8.14). The force on atom n will now be given by

$$F_n = -J(2u_n - u_{n+1} - u_{n-1}) = -4Ju_n = m \frac{\partial^2 u_n}{\partial t^2} \quad (8.16)$$

We can write the displacement of atom n for this wave as

$$u_n = \tilde{u} \exp(i\omega t) \quad (8.17)$$

Substitution of this equation into both sides of eqn 8.16 gives

$$\begin{aligned} -4Ju_n &= -m\omega^2 u_n \\ \Rightarrow \omega &= \left(\frac{4J}{m}\right)^{1/2} \end{aligned} \quad (8.18)$$

The result for $k = \pi/a$ is compared with the result for $k \rightarrow 0$ in Fig. 8.2. It should be noted that we expect that the vibration with $\lambda = 2a$ will be a standing wave, which will have the group velocity $\partial\omega/\partial k = 0$. The interpolation between the results for $k \rightarrow 0$ and $k = \pi/a$ will now be developed exactly using the same methods.

8.3.5 Vibrations with shorter wavelengths: the general case

If we consider each wave vector separately, we can solve the equation of motion for each individual wave propagating along the one-dimensional chain of atoms. We substitute the general equations for u_n and $u_{n\pm 1}$, eqns 8.13 and 8.14, into the force component of eqn 8.12 to obtain the force acting on atom n :

$$\begin{aligned} -\frac{\partial E_n}{\partial u_n} &= -J(2u_n - u_{n+1} - u_{n-1}) \\ &= -Ju_n[2 - \exp(ika) - \exp(-ika)] \\ &= -2Ju_n[1 - \cos(ka)] \\ &= -4Ju_n \sin^2(ka/2) \end{aligned} \quad (8.19)$$

The *mass $\times$ acceleration* term is given as

$$m \frac{\partial^2 u_n}{\partial t^2} = -m\omega_k^2 u_n \quad (8.20)$$

When we equate the two expressions, and divide out u_n from both sides of the equation, we obtain the following expression for the angular frequency as a function of wave vector, ω_k :

$$\begin{aligned} \omega_k^2 &= \left(\frac{4J}{m}\right) \sin^2(ka/2) \\ \Rightarrow \omega_k &= \left(\frac{4J}{m}\right)^{1/2} |\sin(ka/2)| \end{aligned} \quad (8.21)$$

By taking the limit $k \rightarrow 0$ we recover eqn 8.10. By taking only the positive roots we obtain the behaviour of the angular frequency shown in Fig. 8.3. The graph of ω_k is known as a **dispersion curve**. It should be noted that there is a periodicity of the results in k . For every k equal to a reciprocal lattice vector, i.e. $k = 2n\pi/a$, where n is any integer, the angular frequency falls to zero. This result was anticipated in Section 4.5, where we noted that any two waves with wave vectors that differ by a reciprocal lattice vector will be equivalent. It is a rather elegant point to note that we have deduced the existence of the reciprocal

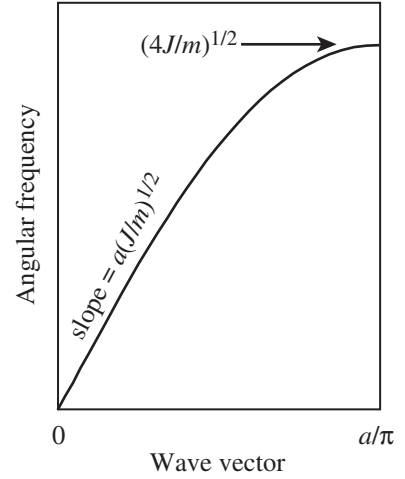


Fig. 8.2 Interpolation of the dependence of frequency on wave vector for the one-dimensional harmonic chain.

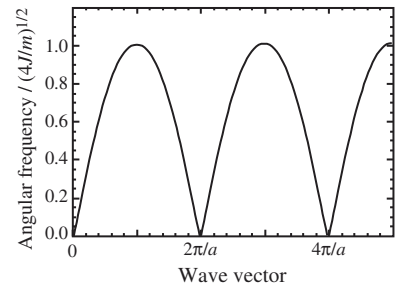


Fig. 8.3 Acoustic mode dispersion curve for the one-dimensional chain.

lattice (at least in one dimension, although the result can be generalized to three dimensions) without having introduced the idea of the reciprocal lattice at any stage in the development of the theory. It persuades this author that reciprocal space is as real a concept as real space, and not, as supposed by some, a creation of crystallographers to make their tasks easier but which is best avoided by everyone else whenever possible!

We note at this point that vibrations with dispersion curves of the form of Fig. 8.3, namely with $\omega \rightarrow 0$ as $k \rightarrow 0$, are called **acoustic modes**. This name highlights the link to sound waves in the limit $k \rightarrow 0$. In Section 8.5 we will introduce another type of vibration, called an **optic mode**, which has a quite different behaviour in the limit $k \rightarrow 0$.

8.3.6 Extension of model of monatomic chain to include distant neighbours

The analysis of the lattice vibrations of the monatomic chain can be extended to include more interactions beyond the nearest neighbour interaction. If $\phi_p(r_p)$ is the energy between the n -th atom and its p -th neighbour, which have equilibrium separation $r_p = pa$, the energy of the chain is given by

$$E = N \sum_p \phi_p(pa) + \frac{1}{2} \sum_{n,p} J_p (u_n - u_{n+p})^2 \quad (8.22)$$

where

$$J_p = \frac{\partial^2 \phi_p}{\partial r_p^2} \quad (8.23)$$

The equation of motion for any atom given by eqn 8.12 is now generalized to

$$m \frac{\partial^2 u_n}{\partial t^2} = -\frac{\partial E}{\partial u_n} = -\sum_p (2u_n - u_{n+p} - u_{n-p}) \quad (8.24)$$

We again substitute the travelling wave solution for each wave vector, eqn 8.13 (and the generalization of eqn 8.14):

$$-m\omega_k^2 u_n = u_n \sum_p J_p [2 - \exp(ikpa) - \exp(-ikpa)] \quad (8.25)$$

As before, this simply reduces to

$$\begin{aligned} m\omega_k^2 &= 2 \sum_p J_p [1 - \cos(kpa)] \\ \Rightarrow \omega_k^2 &= \frac{4}{m} \sum_p J_p \sin^2(kpa/2) \end{aligned} \quad (8.26)$$

This result represents only a minor modification to the result for nearest neighbours. In particular, it has the same behaviour for $k \rightarrow 0$ and at $k = \pi/a$ as described earlier. The main difference is that the additional terms give the possibility that the dispersion curve has a more detailed shape than the simpler nearest-neighbour model.

8.3.7 Reciprocal lattice, the Brillouin zone, and allowed wave vectors

We now recall the general result obtained in Section 4.5 that two waves with wave vectors k and k' that differ by a reciprocal lattice vector G , i.e. $k' = k + G$, have the same wave properties. This was seen in the calculation of the dispersion curves, where the plot of ω_k is periodic in $2\pi/a$. To reiterate the point, we compare the displacements u_{nk} and $u_{nk'}$, eqn (8.13). Noting that $\omega_k = \omega_{k'}$, we rewrite $u_{nk'}$ as

$$\begin{aligned} u(x) &= \tilde{u} \exp(i[k'na - \omega_{k'}t]) \\ &= \tilde{u} \exp(i[(k + G)na - \omega_k t]) \\ &= \tilde{u} \exp(iGna) \exp(i[kna - \omega_k t]) \end{aligned} \quad (8.27)$$

Since $\exp(iGa) = 1$, we see that $u_{nk'} = u_{nk}$, i.e. the two travelling waves have identical effects on the atoms in the chain. This is illustrated in Fig. 8.4, where we see that two waves with wave vectors differing by a reciprocal lattice vector give identical displacements at each lattice position. This result can be generalized to three dimensions, and to unit cells containing more than one atom.

Since the equations of motion for waves with wave vectors that differ by a reciprocal lattice vector are equivalent, the only unique information contained in the dispersion curves is for the range of wave vectors lying between the limits $-\pi/a < k < \pi/a$. This range of wave vectors is the one-dimensional equivalent of the Brillouin zone introduced in Section 4.5. The wave vectors $k = \pm\pi/a$ define special points in reciprocal space, called **Brillouin zone boundaries**, which lie half-way between reciprocal lattice points. The group velocity, $\partial\omega/\partial k$, is equal to zero at these points, meaning that the wave with this wave vector is a standing wave. It will turn out that this is a general result in three dimensions.

It is common practice to represent the wave vector as a normalized quantity by dividing its value by the magnitude of the first reciprocal lattice vector lying along the direction of the wave vector. This gives what is called the **reduced wave vector**. For our one-dimensional example, the wave vector at the Brillouin zone boundary, $k = \pi/a$, is represented by the reduced wave vector of value $1/2$, obtained by dividing the wave vector $a^*/2$ by the reciprocal lattice vector a^* . This is really no different to the normal use of Miller indices.

Although the dispersion curve is a continuous curve, a finite chain will only permit a discrete set of wave vectors. This set is determined by noting that in practice we never define an origin of the crystal, and our equations do

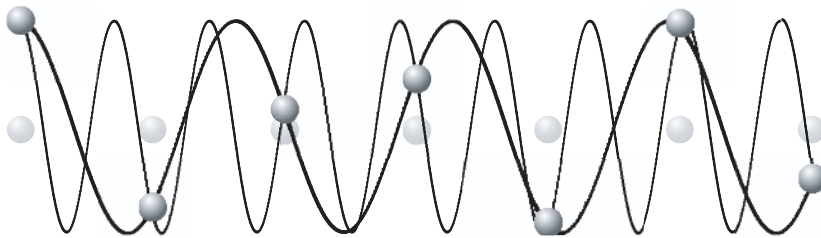


Fig. 8.4 The atomic motions associated with two waves whose wave vectors differ by a reciprocal lattice vector. We see that the two waves have identical effects on the atoms.

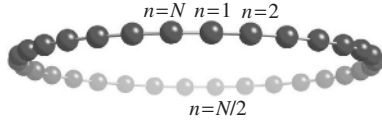


Fig. 8.5 Joined chain to show operation of periodic boundary conditions.

not suppose the existence of any boundaries. In effect, the two ends of the one-dimensional chain of atoms are effectively joined together. This state of having removed the free boundaries by assuming that the periodicity of the structure spans the boundaries in both directions is known as **periodic boundary conditions**. This construction of the periodic boundary conditions is illustrated in Fig. 8.5. We briefly met this construction in Section 5.5.5.

With periodic boundary conditions, the atom at one end of the chain, labelled by the integer $n = N$ with position $x = Na$, is identical to the atom at the origin, which is labelled as $n = 0$ and position $x = 0$. Thus the wave equation gives terms of the form

$$\exp(ikNa) = \exp(0) = 1 \quad (8.28)$$

Therefore the allowed values of k are given by the discrete set:

$$k = \frac{2\pi n}{Na} \quad (8.29)$$

where n is an integer. This shows that there are N allowed wave vectors within one Brillouin zone. In the general case, including the generalization to three dimensions, the number of allowed wave vectors within one Brillouin zone is equal to the number of primitive unit cells in the crystal.

8.3.8 Three-dimensional monatomic crystals: general principles

The one-dimensional model can be easily extended to three dimensions if, in the equations of motions, the variables u refer not to displacements of atoms but of planes of atoms, and appropriate force constants are defined. We also need to include motions that are perpendicular to the wave vector. These follow as a simple extension of our one-dimensional model. The perpendicular displacements will have different force constants from the longitudinal force constants, but they will be expressed in an identical manner so that the results for the different types of motion will follow identical equations. This situation is illustrated in Fig. 8.6.

As there are two orthogonal directions perpendicular to the wave vector, there will be two more acoustic branches with different dispersion curves determined by the new force constants. The corresponding vibrations are called **transverse acoustic modes** (TA), whereas the vibrations where the atoms move in directions parallel to the wave vector are called **longitudinal acoustic modes** (LA). In high symmetry situations, where there are three-, four- or six-fold rotation axes along the direction of the wave vector, the force constants will be the same for any two transverse displacements, so that the two transverse modes will have degenerate (i.e., equal) frequency dispersion.

Experimental data of dispersion curves are commonly represented using the **reduced wave vector scheme** introduced in the previous section. In three dimensions, any general wave vector can be written as

$$\mathbf{k} = \xi \mathbf{a}^* + \zeta \mathbf{b}^* + \varsigma \mathbf{c}^* \quad (8.30)$$

where ξ, ζ, ς typically have values between $\pm 1/2$ (except for non-primitive lattices). The set (ξ, ζ, ς) is directly analogous to the set (h, k, l) used to

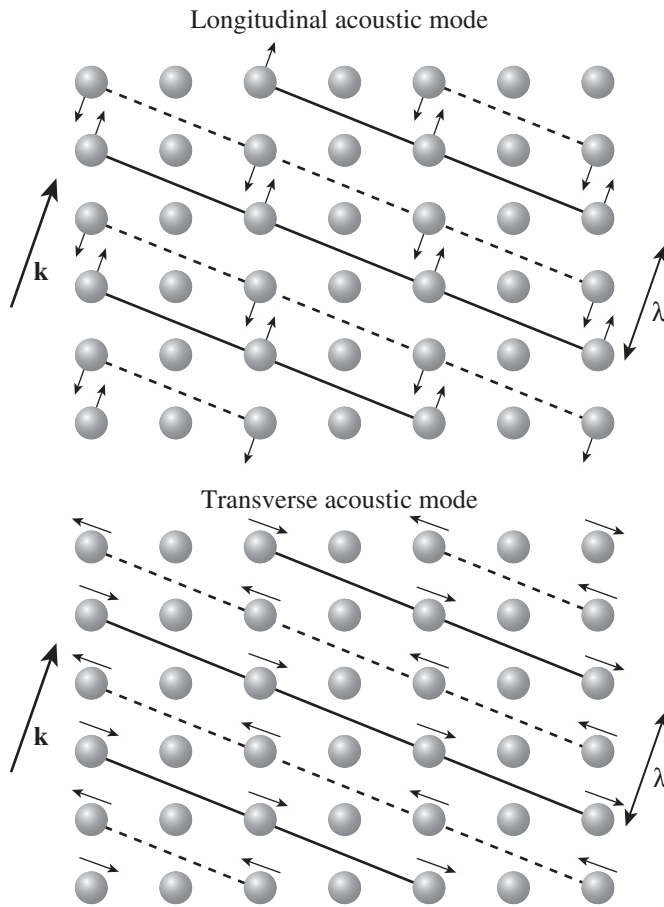


Fig. 8.6 Travelling waves with longitudinal and transverse atomic displacements in a two-dimensional section of a monatomic crystal.

represent lattice planes (Section 4.2), except that the values of ξ , ζ , ς are not confined to integers. The reduced wave vector representation (ξ , ζ , ς) is widely used for representing wave vectors in diagrams of dispersion curves.

8.4 Dispersion curves in face-centred cubic materials

8.4.1 Dispersion curves of neon

The simplest face-centred cubic materials are crystals of the rare gases. The bonding in these materials is primarily of van der Waal's type (Section 5.4.4), and the interactions between second-nearest neighbour atoms are much weaker than those between nearest neighbours. The dispersion curves of crystalline neon are shown in Fig. 8.7. The data are shown using the reduced zone scheme, restricting the range of wave vectors to within the first Brillouin zone. The data for the three symmetry directions are plotted back-to-back to highlight the fact that the Brillouin zone boundaries along the directions $[100]$ and

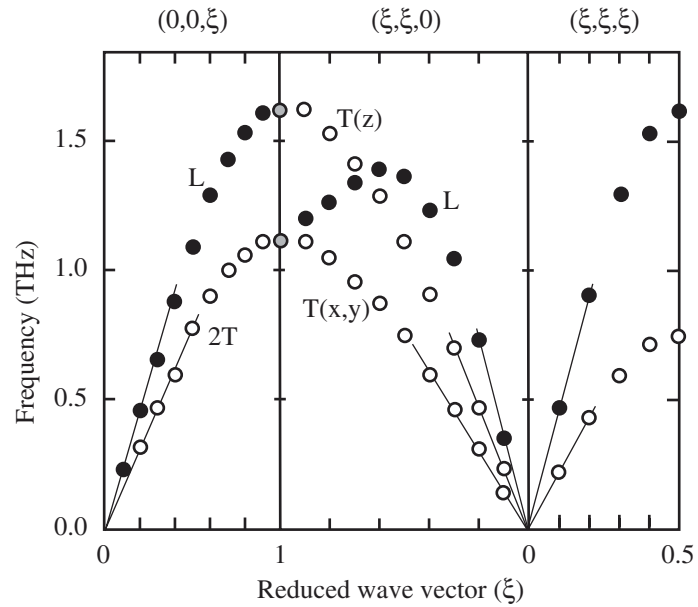


Fig. 8.7 Acoustic mode dispersion curves for neon (ccp structure) measured by inelastic neutron scattering. L and T denote longitudinal and transverse modes respectively. For the [001] and [111] directions the two transverse modes are degenerate (as indicated by the label 2T). Along the [110] direction, the (x,y) and (z) labels indicate the directions of the atomic motions. (Data taken from Endoh *et al.*, *Phys. Rev. B* **11**, 1681, 1975.)

[110] are symmetrically equivalent points in reciprocal space (see below, and Problem 4.5).

There are a number of overall features that should be noted at the outset. The transverse acoustic modes for wave vectors along [100] and [111] are degenerate as a result of the 4-fold and 3-fold rotational symmetry along these directions respectively. On the other hand, the [110] direction contains only a 2-fold rotation axis, so the two transverse modes have different frequencies. The dispersion curves along the [100] and [111] directions closely follow the sinusoidal relationship developed for the simple monatomic chain with nearest-neighbour interactions. The more complex shapes of the dispersion curves along [110] arise from second-neighbour interactions between planes of atoms which arise even when there are only interactions between nearest-neighbour atoms. This will be discussed in more detail below.

Another point that should be noted is that the [100] and [110] points in reciprocal space are symmetrically equivalent (see Problem 4.5), and this means that the dispersion curves along the two directions converge to common frequencies at this point. In particular, it is necessary for two of the modes along [110] to become degenerate at the Brillouin zone boundary. The reason why it is the longitudinal and one of the transverse modes that become degenerate is explored in Problem 8.1.

The data in Fig. 8.7, as in other plots of dispersion curves that will be given later in this chapter, were obtained by inelastic neutron scattering experiments. The basic theory of the method of inelastic neutron scattering will be discussed in Chapter 10, but for the moment we can outline the basic idea. Neutrons are scattered by quanta of lattice vibrations (to be discussed in Chapter 9), and the frequency and wave vector of any vibration can be obtained by measuring the change in energy and wave vector of the scattered neutron beam. The change

in wave vector corresponds to the wave vector of the lattice vibration, and the change in energy is equal to the frequency of the vibration multiplied by Planck's constant.

The dispersion curves of the other rare-gas crystals are very similar to those of neon (these are gathered together in Dove 1993), except that there is a scaling of the frequency range due to differences in mass and in the values of nearest-neighbour force constants.

Dispersion curves of neon along [100]

We can interpret the dispersion curves of neon along [100] by mapping the interatomic interactions onto the interactions between planes. This mapping is shown in Fig. 8.8. The atomic force constants act along the direction of the distance between atoms, and are given by $\partial^2\phi/\partial r^2$, where r is the interatomic distance. The interatomic force constants need to be mapped onto the force constant between planes for both longitudinal and transverse displacements of planes. It can be shown (Problem 8.2) that if we write the interatomic vector as $\mathbf{r} = (x, y, z)$,

$$\frac{\partial^2\phi}{\partial x^2} = \frac{x^2}{r^2} \frac{\partial^2\phi}{\partial r^2} \quad (8.31)$$

Similar results hold for the y and z directions. To obtain the force constants for the three relative displacements of neighbouring planes we need to sum over all interatomic interactions between the planes. If we denote $K = \partial^2\phi/\partial r^2$ we have the following three force constants (Problem 8.3).

$$J_x = \sum_j \frac{\partial^2\phi}{\partial x_j^2} = 2K \quad J_y = \sum_j \frac{\partial^2\phi}{\partial y_j^2} = K \quad J_z = \sum_j \frac{\partial^2\phi}{\partial z_j^2} = K \quad (8.32)$$

J_x gives the force constant for the longitudinal modes, and J_y and J_z give the force constants for the transverse modes. The four nearest-neighbour interatomic vectors between the planes shown in Fig. 8.8 are

$$\frac{a}{2}[1, \pm 1, 0] \quad \frac{a}{2}[1, 0, \pm 1]$$

There are no interactions between second-neighbour planes. We therefore expect both the dispersion curves for the longitudinal and transverse modes

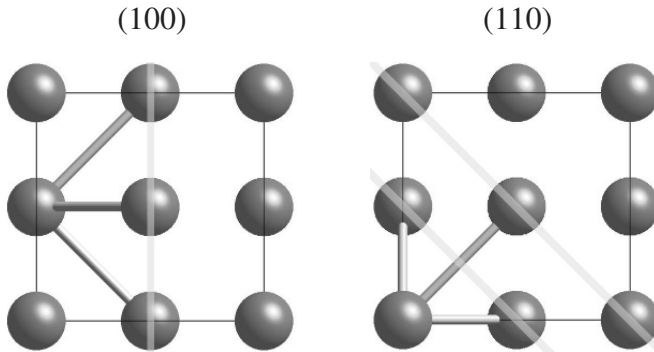


Fig. 8.8 Mapping of interatomic interactions onto the (100) and (110) planes of the ccp crystal, with the planes marked by the thick light lines. The relevant nearest-neighbour distances are represented by the tie lines between atoms. For neighbouring (100) planes there are four interatomic interactions (noting that one lies directly above another in the view of the diagram). For nearest-neighbour (110) planes there are again four interatomic interactions (two being masked out), and there is one interatomic interaction between second-neighbour planes.

along [100] to have the sinusoidal form from before, with the frequency of the longitudinal mode at each wave vector to be $\sqrt{2}$ times larger than the frequencies of the two transverse modes at the same wave vector. Moreover, the frequency of the longitudinal mode should be the same as the LA mode along [100]. Both these features are confirmed by the data shown in Fig. 8.7.

Dispersion curves of neon along [111]

The planes in this case are the close-packed planes, and each atom in one plane interacts with three atoms in the neighbouring plane. It is convenient to define a new coordinate system with $x \parallel [111]$ and $y \parallel [1\bar{1}0]$. This gives the interplanar force constants (Problem 8.3)

$$J_x = 2K \quad J_y = J_z = K/2 \quad (8.33)$$

Again, there are no interactions between second-neighbour planes. The ratio of the two force constants suggest that the frequency of the longitudinal mode should be twice as large as the frequencies of the corresponding transverse modes for wave vectors along [110], and this is confirmed by the data in Fig. 8.7.

Dispersion curves of neon along [110]

The mapping of the atomic model onto the (110) planes is shown in Fig. 8.8. It is now convenient to define a coordinate system such that $x \parallel [110]$, $y \parallel [1\bar{1}0]$, and $z \parallel [001]$. In this case, the mapping shows that there is an interaction between second-nearest planes. The mapping gives the following force constants for nearest-neighbour planes (Problem 8.3):

$$J_x^{(1)} = J_y^{(1)} = K \quad J_z^{(1)} = 2 \frac{\partial^2 \phi}{\partial z^2} = 2K \quad (8.34)$$

and the following force constants for second-neighbour planes:

$$J_x^{(2)} = K \quad J_y^{(2)} = J_z^{(2)} = 0$$

In this case the dispersion curves for the two transverse modes will be the standard sinusoidal form, with one of the modes having frequencies that are $\sqrt{2}$ times larger than the frequencies of the second TA mode. The LA mode will have the extended form given by eqn 8.26, involving the sum of two sinusoidal contributions to ω^2 . The second contribution will have a maximum at the wave vector half-way to the Brillouin zone boundary, and zero contribution at the zone boundary. Therefore we expect the LA mode to have the same value of the frequency at the zone boundary as the TA(y) mode. These features are confirmed in the data of Fig. 8.7.

We have therefore been able to rationalize the form of the dispersion curves of crystalline neon for the three principal symmetry directions in reciprocal space. The good agreement between the simple theory and the experimental data shows that taking only nearest-neighbour interatomic interactions is a very good approximation. The shape of the LA curve along [110] shows that mapping a real three-dimensional crystal onto the linear chain model may lead to second-neighbour interplanar interactions even though the interatomic interactions are only taken to nearest-neighbour pairs.

8.4.2 Dispersion curves of lead

Crystalline lead also has a fcc structure. The measured dispersion curves are shown in Fig. 8.9. These show many of the features seen in the dispersion curves of the crystalline rare gas elements. However, it is clear from the shapes of the curves along $[100]$ and $[110]$ that there are important interatomic interactions that extend beyond nearest-neighbour pairs of atoms. There are also glitches in the continuous dispersion curves, most obviously in the LA mode along $[110]$ at the wave vector around $[0.3, 0.3, 0]$, which arise when the wave vectors match the wave vectors of the electronic Fermi surface (Section 5.4.7).

8.4.3 Dispersion curves of potassium

Crystalline potassium has a body-centred cubic structure. The dispersion curves are shown in Fig. 8.10. In the bcc lattice the Brillouin zone boundaries along the $[100]$ and $[111]$ directions are symmetrically equivalent points in reciprocal space, and this is highlighted in the scheme chosen to represent the dispersion curves.

The forms of the LA and TA dispersion curves along $[100]$ and of the TA mode along $[110]$ show that the nearest-neighbour interatomic interactions dominate. The mapping of the interatomic interactions onto the interactions between planes is shown in Fig. 8.11. The vectors between an atom in one plane and its four neighbours in the first-nearest (100) plane are

$$\frac{a}{2}(1, \pm 1, \pm 1)$$

It can be shown (Problem 8.6) that all force constants are equal:

$$J_x = \sum_j \frac{\partial^2 \phi}{\partial x_j^2} = 2K \quad J_y = \sum_j \frac{\partial^2 \phi}{\partial y_j^2} = 2K \quad J_z = \sum_j \frac{\partial^2 \phi}{\partial z_j^2} = 2K$$

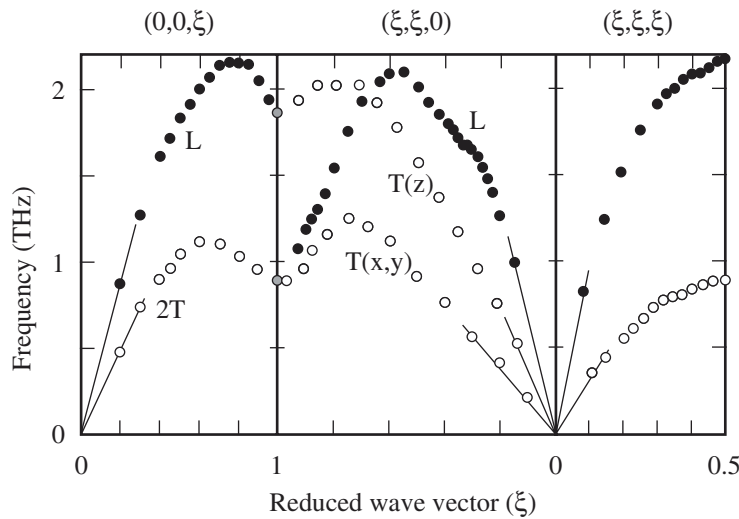


Fig. 8.9 Acoustic mode dispersion curves for lead (fcc) measured by inelastic neutron scattering. (Data taken from Brockhouse *et al.*, *Phys. Rev.* **128**, 1099, 1962.)

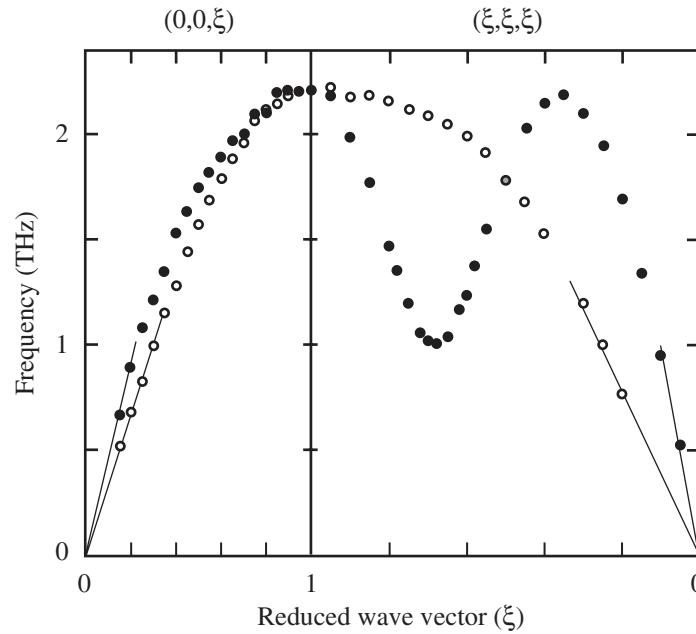


Fig. 8.10 Acoustic mode dispersion curves for potassium (bcc) measured by inelastic neutron scattering. The right-hand plot shows the relevant portion of reciprocal space. (Data taken from Cowley *et al.*, *Phys. Rev.* **150**, 487, 1966.)

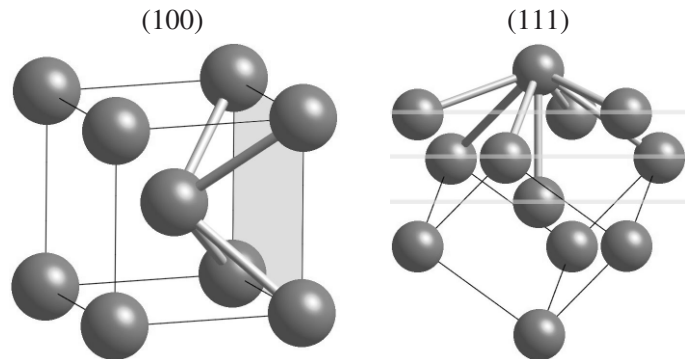


Fig. 8.11 Mapping of interatomic interactions onto the (100) and (111) planes of the bcc crystal, with the planes marked by the light shading. The nearest-neighbour interactions are represented by the tie lines between atoms. There are four interatomic interactions contributing to the (100) interplanar force constants. On the other hand, for the (111) planes, the nearest-neighbour interatomic interactions contribute to interplanar interactions up to third-distant neighbouring planes.

This shows why, if the nearest-neighbour interatomic interactions dominate, the LA and TA modes have very similar frequencies for wave vectors along [100], as seen in Fig. 8.10. The second-neighbour interatomic interaction involves the separation of one unit cell length, and this will contribute to only the LA interplanar force constant. The small contribution of this second-neighbour interaction can be seen in the small increase in frequency of the LA mode over the TA mode that reaches a maximum at the reduced wave vector $(\frac{1}{2}, 0, 0)$.

The interactions between (111) planes are a little more complicated, as seen in Fig. 8.11. The interactions involved in the nearest-neighbour planes are the nearest-neighbour interatomic distances, but the projection is at a wide angle, meaning that the contribution to the longitudinal force constants is not large. The interaction between second-neighbour planes involves second-neighbour interatomic interactions, and these are expected to be somewhat weaker. The

third-neighbour interplanar force constants also involve the nearest-neighbour interatomic interaction. In this case the interatomic vector is normal to the planes, so there will be a large contribution to the longitudinal force constant but no interaction to the transverse force constants. If we define K as the first-neighbour interatomic force constant as in our analysis of the ccp structures, and set the second-neighbour interatomic force constant to zero, it can be shown that the interplanar force constants are

$$J_L^{(1)} = K/3 \quad J_T^{(1)} = 4K/3 \quad J_L^{(3)} = K$$

$J_L^{(2)}$ is only sensitive to second-neighbour interactions, and is therefore likely to be lower in value than $J_L^{(1)}$. Clearly $J_L^{(3)}$ is the largest of the longitudinal force constants. Writing d as the (111) interplanar distance, and projecting the wave vector onto the [111] direction, the LA dispersion curve has the form

$$\omega_{LA}^2 = \frac{4}{m} \left(J_L^{(1)} \sin^2(kd/2) + J_L^{(2)} \sin^2(kd) + J_L^{(3)} \sin^2(3kd/2) \right) \quad (8.35)$$

As we noted, the strongest force constant is $J_L^{(3)}$, and it can be seen that the contribution from this force constant vanishes when $k = 2\pi/3d$, which corresponds to the three-dimensional reduced wave vector $[\frac{2}{3}, \frac{2}{3}, \frac{2}{3}]$. This is the origin of the large dip in the dispersion curve at this wave vector seen in the data of Fig. 8.10 (see Problem 8.7).

8.5 Lattice vibrations of crystals with several atoms in the unit cell

8.5.1 The basic model

The main features of the lattice dynamics of crystals with several atoms in the unit cell can be represented by a simple model that is an extension of the one-dimensional chain. This is shown in Fig. 8.12. Two types of atoms of masses M and m interact with force constant J , and their displacements U and u are defined in this figure. All atoms are separated by $a/2$ when at rest. This model represents crystals that have alternating planes of two types of atoms, as for the (111) planes of atoms in the NaCl structure. We will consider only the longitudinal motions, as the extension to include transverse motions is the same as for the monatomic case. We will also only consider nearest-neighbour interactions. This model will give us some new important characteristics, but despite its simplicity it proves to be about as far as we can go before the

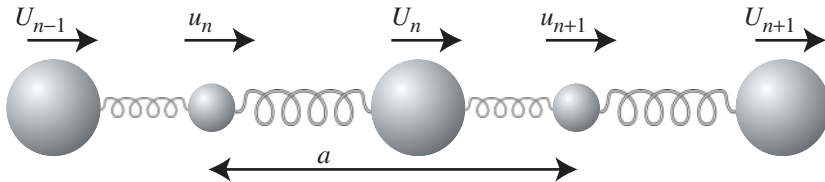


Fig. 8.12 Diatomic linear harmonic chain. The model is shown here with two distinct types of atoms and two distinct harmonic interactions. In the main text we treat the two types of interactions as equivalent, and Problem 8.11 treats the case of distinct interactions and both atoms having the same mass.

arithmetic becomes too complicated. For anything more complex we will need to resort to the computer!

The harmonic energy of the model is given as

$$E = \frac{1}{2} J \sum_n (U_n - u_n)^2 + (u_{n+1} - U_n)^2 \quad (8.36)$$

Following the procedure used for the analysis of the monatomic chain, we can write down the equations for the forces acting on both types of atoms. For the big atom labelled n , the energy is given as

$$E(U_n) = \frac{1}{2} J [(U_n - u_n)^2 + (u_{n+1} - U_n)^2] \quad (8.37)$$

and the force on it is given by

$$\begin{aligned} F_n &= -\frac{\partial E(U_n)}{\partial U_n} = -J[(U_n - u_n) - (u_{n+1} - U_n)] \\ &= -J(2U_n - u_n - u_{n+1}) \\ &= M \frac{\partial^2 U_n}{\partial t^2} \end{aligned} \quad (8.38)$$

Similarly, for the small atom labelled n , the energy is given as

$$E(u_n) = \frac{1}{2} J [(u_n - U_{n-1})^2 + (U_n - u_n)^2] \quad (8.39)$$

and the force on it is given by

$$\begin{aligned} f_n &= -\frac{\partial E(u_n)}{\partial u_n} = -J[(u_n - U_{n-1}) - (U_n - u_n)] \\ &= -J(2u_n - U_{n-1} - U_n) \\ &= m \frac{\partial^2 u_n}{\partial t^2} \end{aligned} \quad (8.40)$$

We can assume the same general solutions as used for the monatomic chain, one for each atom type:

$$\begin{aligned} U_n &= \sum_k \tilde{U}_k \exp(i[kna - \omega_k t]) \\ u_n &= \sum_k \tilde{u}_k \exp(i[kna - \omega_k t]) \end{aligned} \quad (8.41)$$

where k is the wave vector, ω_k is the angular frequency of a given mode, and $\tilde{U}_k$ and $\tilde{u}_k$ are the two amplitudes for a single given mode (the two atoms each have a different amplitude for any one mode).

8.5.2 Solution for small wave vector

Consider the two types of waves illustrated in Fig. 8.13. The first wave is a sound wave, where all atoms move in phase. An extension of the earlier model leads to the result for the acoustic mode frequency for small k :

$$\omega = a \left(\frac{J}{2(M + m)} \right)^{1/2} k \quad (8.42)$$

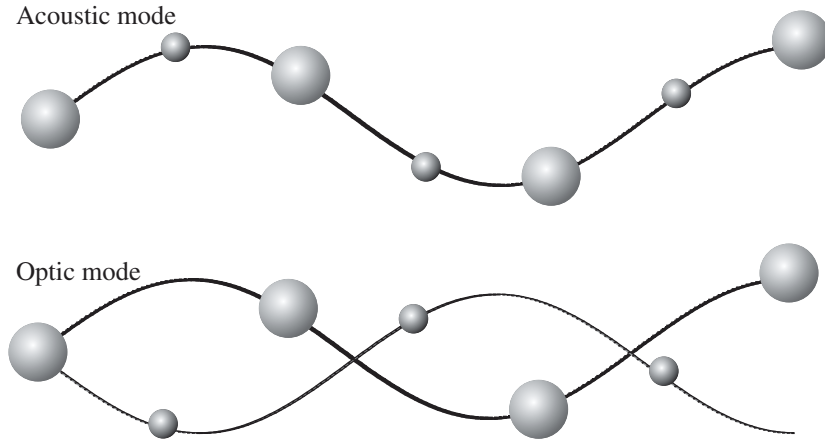


Fig. 8.13 Exaggerated transverse atomic motions in a one-dimensional diatomic crystal for an acoustic mode (motions of different types of atoms are in phase) and an optic mode (motions of different types of atoms are out of phase).

The second wave has neighbouring atoms (of different types) moving out of phase. For ionic crystals, this wave will have the cations moving in opposite directions to neighbouring anions. For $k = 0$ (infinite wavelength) the condition that the centre of mass doesn't move means that

$$MU_n = -mu_n \quad (8.43)$$

As a result we can write

$$u_n = \frac{A}{m} \exp(i\omega t) \quad U_n = -\frac{A}{M} \exp(i\omega t) \quad (8.44)$$

Note that at $k = 0$, $u_n = u_{n+1}$ and $U_n = U_{n+1}$ for all n . We substitute these solutions into the equation of motion for the big atom:

$$\begin{aligned} M \frac{\partial^2 U_n}{\partial t^2} &= -A\omega^2 \exp(i\omega t) = -\frac{\partial E}{\partial U_n} \\ &= -J(U_n - u_{n+1} - u_n) \\ &= -J \left(2\frac{A}{M} + \frac{A}{m} + \frac{A}{m} \right) \exp(i\omega t) \\ &= -2AJ \left(\frac{1}{M} + \frac{1}{m} \right) \exp(i\omega t) \end{aligned} \quad (8.45)$$

This leads to the result for the angular frequency:

$$\omega^2 = 2J \left(\frac{1}{M} + \frac{1}{m} \right) \quad (8.46)$$

The same result is obtained when we repeat the procedure for the small atom (Problem 8.9). Unlike the acoustic mode, this vibration has a non-zero frequency at $k = 0$ which is, to first order, independent of k . This is a *standing wave*, which is properly defined as such through the condition $\partial\omega/\partial k = 0$. This is called the **optic mode**, because if the two atoms are of opposite charge their relative motions correspond to the motions that would be induced by the passage of an electromagnetic wave of optical frequencies. Actually the frequencies of optic modes are usually within the infrared region of the electromagnetic spectrum, as we will see in the examples later in this chapter.

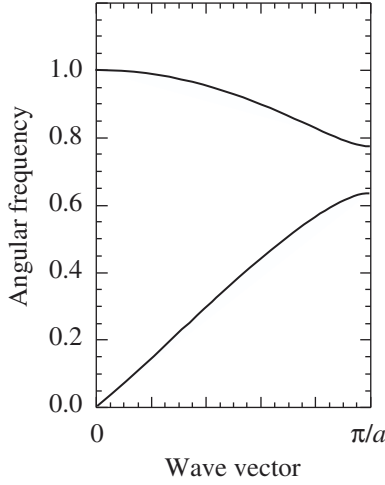


Fig. 8.14 Dispersion curves for the one-dimensional diatomic chain obtained from eqn 8.49.

8.5.3 General result

The general solution for all wave vectors can be obtained with a little more algebra. Using eqns 8.37–8.41, we have

$$\begin{aligned} -M\omega^2\tilde{U} &= -J\{2\tilde{U} - \tilde{u}[1 + \exp(ika)]\} \\ -m\omega^2\tilde{u} &= -J\{2\tilde{u} - \tilde{U}[1 + \exp(-ika)]\} \end{aligned} \quad (8.47)$$

These equations can be rewritten in matrix form:

$$\begin{pmatrix} M\omega^2 - 2J & J[1 + \exp(ika)] \\ J[1 + \exp(-ika)] & m\omega^2 - 2J \end{pmatrix} \begin{pmatrix} \tilde{U} \\ \tilde{u} \end{pmatrix} = 0 \quad (8.48)$$

This solution of this matrix equation is that for which the determinant of the matrix is zero. This gives

$$\begin{aligned} [M\omega^2 - 2J][m\omega^2 - 2J] &= J^2[1 + \exp(ika)][1 + \exp(-ika)] \\ \Rightarrow Mm\omega^4 - 2J(M + m)\omega^2 + 4J^2 \sin^2(ka/2) &= 0 \\ \Rightarrow \omega^2 &= J \frac{M + m}{Mm} \pm \frac{J}{Mn} \sqrt{(M + m)^2 - 4Mm \sin^2(ka/2)} \end{aligned} \quad (8.49)$$

This general result is shown for one set of masses in Fig. 8.14, which shows both the acoustic and optic mode frequencies as functions of the wave vector. It can readily be shown (Problem 8.10) that the results given earlier for $k \rightarrow 0$ (eqns 8.42 and 8.46) are obtained from this general result. The general result also allows us to solve for the atomic motions. For the two solutions for $k \rightarrow 0$ it is straightforward to show that the acoustic mode solution for the frequency gives $\tilde{U} = \tilde{u}$, and that the optic model solution gives $M\tilde{U} = -m\tilde{u}$.

It is interesting to consider the behaviour at the Brillouin zone boundary, $k = \pi/a$. In this case the two frequencies can be shown to be (Problem 8.10)

$$\omega^2 = \frac{2J}{M} \quad \omega^2 = \frac{2J}{m} \quad (8.50)$$

The matrix equation (eqn 8.48) then simplifies to

$$\begin{pmatrix} M\omega^2 - 2J & 0 \\ 0 & m\omega^2 - 2J \end{pmatrix} \begin{pmatrix} \tilde{U} \\ \tilde{u} \end{pmatrix} = 0 \quad (8.51)$$

For the first solution, $\omega^2 = 2J/M$, $\tilde{u} = 0$ and the value of $\tilde{U}$ is undetermined. For the second solution, $\omega^2 = 2J/m$, we now have $\tilde{U} = 0$ and the value of $\tilde{u}$ is undetermined. So for both modes, one atom moves while the other stands still.

The value of the amplitude of the atom that moves cannot be determined by the dynamic equations developed so far. It will depend on temperature, as will be shown in the next chapter. The fact that in both modes only one atom moves suggests that at the Brillouin zone boundary we have lost the distinction between acoustic and optic modes. This is not always the case (for example, a similar model with two equal masses but two different force constants will retain the difference between acoustic and optic modes in the displacements for all wave vectors, Problem 8.11), but in many crystals the distinction between acoustic and optic modes is only properly maintained in the limit $k \rightarrow 0$. This point is not always appreciated!

8.5.4 Generalization for more complex cases: atomic motions

We reinforce our descriptions of acoustic and optic modes by considering the case of four atoms in a unit cell, all moving along the x direction, in either a positive or negative sense, and with $k \rightarrow 0$. These four motions are shown in Fig. 8.15 (a similar picture is given in Fig. 3.13). There are four combinations of motions of the four atoms, $++++$, $++--$, $+--+$, and $+- -+$, where the signs denote the relative signs of the displacements of the four atoms. These are the only four independent linear combinations of positive or negative displacements, which were labelled according to their irreducible representations in Section 3.4. For the last three combinations, the sums of the displacements are zero, which will therefore correspond to the optic modes. The first mode, for which all the atoms move in phase, is the acoustic mode. All other possible displacement patterns correspond to linear combinations of these basic modes.

It is easy to generalize this argument for Z atoms per cell, showing that for displacements along one direction there will always be one acoustic mode and $(Z - 1)$ optic modes. If we now allow motions along the y and z directions as well, it is clear that we will have three acoustic modes and $(3Z - 3)$ optic modes. We can conclude that the principal difference between acoustic and optic modes is that in the limit $k \rightarrow 0$ the acoustic modes have all the atoms moving in phase, whereas the optic modes have the atoms moving out of phase. This distinction is not relevant for larger wave vectors, as we have seen in one case for our simple one-dimensional model. (Note that if the atoms have different masses, the displacement patterns described above will describe mass-weighted displacements.)

In high-symmetry situations, it is possible to separate **longitudinal optic** (LO) modes from **transverse optic** (TO) modes, just as we noted the separation of LA and TA modes in Section 8.3.8. The distinction is easy to appreciate in simple crystals such as NaCl (Section 8.5.6), but it may not be appropriate in more complex systems.

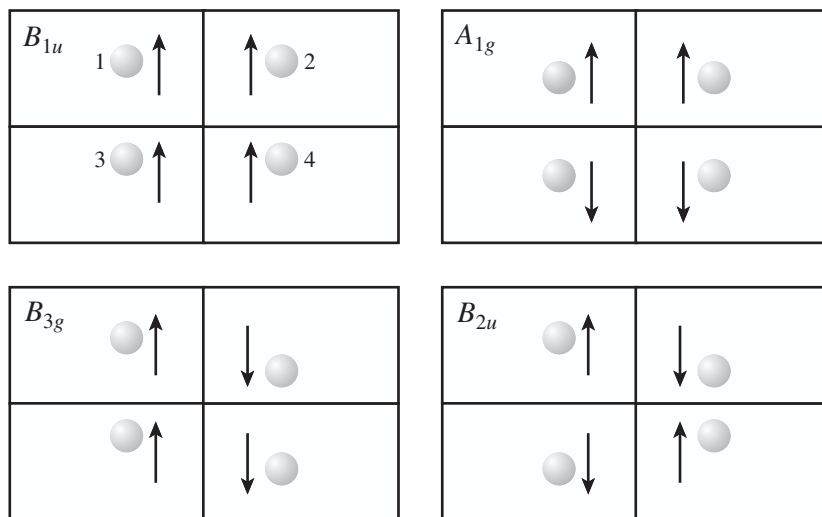


Fig. 8.15 Examples of four modes of motion along x in a two-dimensional crystal of point group symmetry $mm2$ (similar to Fig. 3.13). The model labelled B_{1u} is the acoustic mode; the three other modes are optic modes. For the A_{1g} mode, in which the atomic displacements do not break the symmetry, the atoms are drawn in their mean positions.

8.5.5 Generalization for more complex cases: the dynamical matrix

The mathematical formalism developed for the diatomic chain can be generalized for more complex cases, but only by imposing a certain order on the procedure we used. We will consider another one-dimensional example where we allow interactions to extend beyond nearest neighbour, and we will allow an arbitrary number of atoms in the unit cell. Consider each atom in the unit cell to be labelled j . The energy is now written as

$$E_j = \frac{1}{2} \sum_{n,j'} J_{j,j',n} (u_{j,0} - u_{j',n})^2 \quad (8.52)$$

where the sum over j' runs over all atoms in other unit cells, and n labels the unit cell such that $n = 0$ is the reference unit cell, and the sum over n also includes the $n = 0$ components. The force on the atom is given by

$$\begin{aligned} F_j &= -\frac{\partial E_j}{\partial u_{j,0}} = -\sum_{n,j'} J_{j,j',n} (u_{j,0} - u_{j',n}) \\ &= m_j \frac{\partial^2 u_{j,0}}{\partial t^2} \end{aligned} \quad (8.53)$$

As before, we assume that any wave will affect all atoms in the unit cell. We have a general equation for the displacement of an atom that has the form we have met earlier, but we now need to include the sum over all the modes. Strictly the sum should include all wave vectors, but since this sum will occur in both sides of the force equation we can account for the different wave vectors separately. It will also become convenient to express the atomic displacement equation in a way that separates out a mass term, which may seem to be a little odd for the moment:

$$\begin{aligned} u_{j,n} &= \sum_v \tilde{u}_{j,v} \exp(i[kna - \omega_v t]) \\ &= \sum_v A_v m_j^{-1/2} e_{j,v} \exp(i[kna - \omega_v t]) \end{aligned} \quad (8.54)$$

where v labels the mode (in a one-dimensional system there are as many modes as atoms in the unit cell). A_v is a scale factor that will drop out of the equations – this is the part that is dependent on temperature. The new variables $e_{j,v}$ are normalized such that

$$\sum_j |e_{j,v}|^2 = 1 \quad (8.55)$$

for each value of v . These normalized variables contain the information about the *relative* displacements of the atoms due to the wave.

Substituting the general wave equation into the force equation gives, for each value of v :

$$-m_j^{1/2} e_{j,v} \omega_v^2 = -\sum_{n,j'} J_{j,j',n} \left(m_j^{1/2} e_{j,v} - m_{j'}^{1/2} e_{j',v} \exp(ikna) \right) \quad (8.56)$$

This equation can be written in matrix form. If we define the $N \times 1$ matrix $\mathbf{e}_v$ as formed from the different components $e_{j,v}$, we can write

$$\mathbf{e}_v \omega_v^2 = \mathbf{D} \times \mathbf{e} \quad (8.57)$$

where the $N \times N$ matrix $\mathbf{D}$, which is called the **dynamical matrix**, has components

$$D_{j,j'} = (m_j m_{j'})^{-1/2} \sum_n J_{j,j',n} (\delta_{j,j'} - \exp(ikna)) \quad (8.58)$$

We can pack together the individual $N \times 1$ matrices $\mathbf{e}_v$ to form the $N \times N$ matrix $\mathbf{e}$. This matrix has the property that $\mathbf{e}^{-1} = (\mathbf{e}^*)^T$. We can also form the matrix $\mathbf{\Omega}$ with components $\Omega_{v,v'} = \omega_v^2 \delta_{v,v'}$ (i.e. $\mathbf{\Omega}$ is a diagonal matrix, whose components are the squares of the vibrational angular frequencies). By using these new matrices, we can write

$$\mathbf{e} \times \mathbf{\Omega} = \mathbf{D} \times \mathbf{e} \Rightarrow \mathbf{\Omega} = \mathbf{e}^{-1} \times \mathbf{D} \times \mathbf{e} \quad (8.59)$$

This is now in the form of an **eigenproblem**: the components of the matrix $\mathbf{\Omega}$ are the eigenvalues of $\mathbf{D}$, and the components of $\mathbf{e}$ contain the corresponding eigenvectors. For this reason, the vectors $\mathbf{e}$ are known as the **mode eigenvectors**. Finding the eigenvalues and eigenvectors of a matrix is a task that has been optimized for calculations on a computer. The task then is to set up the dynamical matrix using the individual interatomic force constants, and then find its eigenvalues to get the mode frequencies. Subsuming the masses into the definitions of the $\mathbf{D}$ and $\mathbf{e}$ matrices has allowed the writing of the dynamical equations in the form of an eigenvalue problem.

The dynamical matrix formalism can readily be extended for calculations on three-dimensional crystals. The $\exp(ikna)$ phase factor of the one-dimensional chain can be replaced by $\exp(i\mathbf{k} \cdot \mathbf{r})$, where $\mathbf{r}$ is the vector between the reference unit cell and the neighbouring unit cell, and $\mathbf{k}$ is the three-dimensional wave vector. The other change is that account has to be taken of the three components of the displacement vectors, which means that each component of the dynamical matrix has to be replaced by a 3×3 array representing the combinations of the components of each pair of displacements. Similarly the **frequency matrix** $\mathbf{\Omega}$ and the eigenvector matrix $\mathbf{e}$ both become $3N \times 3N$ matrices.

Finally, we remark that the form of the dynamical matrix will be controlled by the symmetry of the crystal. If we recall the example earlier of four atoms in an orthorhombic crystal of symmetry $mm2$, we can remark that the symmetry of the crystal will ensure that the diagonalization of the dynamical matrix automatically produces the combinations of symmetry that were drawn earlier. We illustrate this by considering the crystal of $mm2$ symmetry shown in Fig. 8.15. We will assume that the atoms have interactions only with nearest neighbours, but a more complicated model will not actually modify the ideas in the analysis. It can be shown (Problem 8.13) that the dynamical matrix at $\mathbf{k} = 0$ has the form

$$\mathbf{D}(\mathbf{k} = 0) = \begin{pmatrix} a & b & c & d \\ b & a & d & c \\ c & d & a & b \\ d & c & b & a \end{pmatrix} \quad (8.60)$$

where b , c , and d involve the second differentials of the interactions between atoms 1 and 2, 1 and 3, and 1 and 4 respectively, and will have negative values. It can be shown that $a = -(b + c + d)$. We consider in turn the four eigenvectors, labelled according to their irreducible representations:

$$\begin{aligned} \mathbf{e}_1(B_{1u}) &= \begin{pmatrix} +1 \\ +1 \\ +1 \\ +1 \end{pmatrix} & \mathbf{e}_2(A_{1g}) &= \begin{pmatrix} +1 \\ +1 \\ -1 \\ -1 \end{pmatrix} \\ \mathbf{e}_3(B_{3g}) &= \begin{pmatrix} +1 \\ -1 \\ +1 \\ -1 \end{pmatrix} & \mathbf{e}_4(B_{2u}) &= \begin{pmatrix} +1 \\ -1 \\ -1 \\ +1 \end{pmatrix} \end{aligned}$$

We now construct the equations of the form $\mathbf{D} \cdot \mathbf{e}_j$, and if the proposed vectors $\mathbf{e}_j$ are true eigenvectors of $\mathbf{D}$ we expect to find $\mathbf{D} \cdot \mathbf{e}_j = \omega_j^2 \mathbf{e}_j$, where ω_j^2 will be a function of the components of $\mathbf{D}$. Indeed, by forming the matrix product $\mathbf{D} \cdot \mathbf{e}_j$ for each j we find that the vectors $\mathbf{e}_j$ are all eigenvectors of $\mathbf{D}$, with

$$\begin{aligned} \omega_1^2 &= a + b + c + d & \omega_2^2 &= a + b - c - d \\ \omega_3^2 &= a - b + c - d & \omega_4^2 &= a - b - c + d \end{aligned}$$

From the condition $a + b + c + d = 0$ noted earlier, we have the result that $\omega_1 = 0$, as expected for the acoustic mode eigenvector when $k \rightarrow 0$.

8.5.6 Lattice dynamics of ionic crystals

The dynamical matrix has two parts. The first is the long-range Coulomb part. This extends to infinite range, and needs to be handled with an Ewald summation as briefly discussed in Section 5.4.1. The second part consists of the short-range interactions. This part can be treated in two ways. On one hand, the individual values of the $J_{j,j',n}$ parameters can be taken as adjustable parameters, tuned to give the best agreement with the experimental measurements of the dispersion curves. On the other hand, the $J_{j,j',n}$ parameters can be written as the differentials of an assumed potential energy function, with a form such as the Buckingham potential (eqn 5.23). In this case the same function can be used to determine values of $J_{j,j',n}$ for different neighbouring unit cells. This has the advantage of tying the force constants to a physical model, which has fewer adjustable parameters which can also be optimized against structural parameters.

Early in the development of lattice dynamics models that were being used to describe the first experimental data on lattice dynamics (the early 1960s) it was realized that it is necessary to take account of the ionic polarizability. This is highlighted by a relationship for simple cubic ionic materials called the **Lyddane-Sachs-Teller (LST) relation**. This relates the limiting values of the LO and TO modes as $k \rightarrow 0$ to the dielectric constants for electric fields of zero and infinite frequency through the simple equality of ratios

$$\frac{\epsilon(\omega = 0)}{\epsilon(\omega = \infty)} = \frac{\omega_{\text{LO}}^2}{\omega_{\text{TO}}^2} \quad (8.61)$$

The frequencies in the ratio are for the LO and TO modes with similar atomic motions. For example, for alkali halides, the LST relation might apply to vibrations with wave vectors of the form $\mathbf{k} = [\eta, 0, 0]$ as $\eta \rightarrow 0$. In this case the LO modes have the cations and anions moving in opposite directions parallel to [100], and the TO modes have the cations and anions moving in opposite directions perpendicular to [100]. Since both types of motion are along symmetrically equivalent directions, when $k = 0$ the types of motion become exactly equivalent. In this case one might have expected that the LO and TO frequencies would be equal. But for wave vectors close to but not exactly at $k = 0$, the fact that the long-wavelength optic modes generate electric fields that are either parallel or perpendicular to the direction of propagation of the optic mode will have a significant effect on the frequency of the mode. Since $\epsilon(\omega = 0) > \epsilon(\omega = \infty)$, it follows that $\omega_{\text{LO}} > \omega_{\text{TO}}$. This effect is known as **LO/TO splitting**, reflecting the fact that at exactly zero wave vector the frequencies are the same. We will see examples of the size of the LO/TO splitting in data for alkali halides below.

The importance of ionic polarizability is that it determines the size of $\epsilon(\omega = \infty)$, which would have a value of 1 without polarizability. It also makes a significant contribution to $\epsilon(\omega = 0)$, which would otherwise only be determined by the polarizability of the crystal structure through displacements of the ions. The shell model introduced in Section 5.4.5 works reasonably well in reproducing experimental values of the two dielectric constants.

The shell model can be incorporated into the formalism of the dynamical matrix by writing separate equations of motion for the core and the shell. Because the mass of the shell is zero in the shell model, there is no separate $m\omega^2$ for the shell. Instead, the shell instantaneously relaxes to its equilibrium position so that there is no restoring force on the shell. Thus the displacement of the shell is completely tied to the displacements of the cores, and therefore there is not a separate solution for the shells. The contribution of the shell folds into an expanded dynamical matrix for the ionic cores, giving the same number of solutions as for models without ionic shells.

8.5.7 The lattice dynamics of the alkali halides

The alkali halides were among the first crystals whose dispersion curves were measured by the emerging technique of inelastic neutron scattering. Measured dispersion curves for NaCl are shown in Fig. 8.16, using the same reduced scheme as shown earlier for the dispersion curves of fcc monatomic crystals. There are two ions in the primitive cell, so there are three acoustic and three optic modes. For wave vectors along the [001] and [111] symmetry directions the TO and TA modes are both degenerate, similar to the degeneracy of the TA modes in argon, lead and potassium (Figs 8.7, 8.9 and 8.10 respectively).

The dispersion curves for the alkali halides show two important features of the lattice dynamics of ionic crystals. First, the data highlight the significance of the LO/TO splitting. In NaCl, the LO and TO modes have frequencies of around 8 and 5 THz respectively, which are significantly different. From eqn 8.61, the ratio of the squares of the LO and TO frequencies suggests that the static dielectric constant is 2.5 times larger than the high-frequency dielectric constant. In fact the experimental values are $\epsilon(\omega = 0) = 5.90$ and $\epsilon(\omega = \infty) = 2.34$, giving, from eqn 8.61, the ratio $\omega^2(\text{LO})/\omega^2(\text{TO}) = 2.52$.

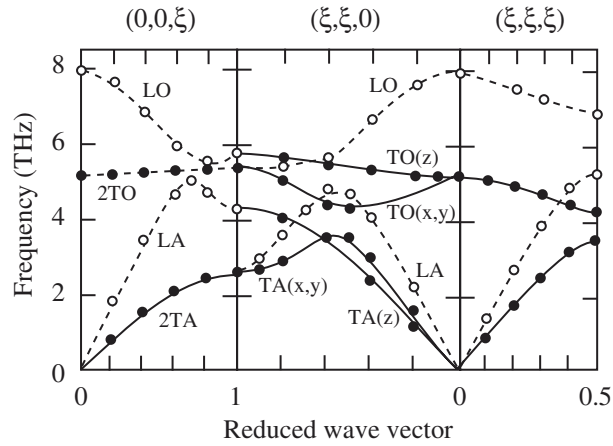


Fig. 8.16 Acoustic and optic mode dispersion curves for NaCl along the main symmetry directions, as measured by inelastic neutron scattering. (Data taken from Raunio *et al.*, *Phys. Rev.* **178**, 1496, 1969.)

A second feature to note is that there is not a separation of optic and acoustic mode frequencies across the range of wave vectors. The simple model of the diatomic chain given earlier showed a separation of the optic and acoustic mode frequencies, but this is an artefact of the simple nature of the model. Consider the two sets of modes for wave vectors along [111]. In this case both the transverse and longitudinal modes show a clear separation of the optic and acoustic modes. In this case there are alternating planes of cations and anions normal to [111] as in the simple diatomic chain, so we might expect the behaviour to resemble the simple model. However, for wave vectors along [100] the planes of atoms normal to this direction contain both cations and anions, so there is much less of a match with the diatomic chain model. The transverse modes do indeed show a separation of the optic and acoustic modes, but there is not a separation of the LO and LA modes. Indeed, for NaCl the dispersion curves for these two types of modes appear to cross. In fact symmetry plays an important role in this case, because the two types of vibration have the same symmetry. Modes with the same symmetry are not permitted to cross in the dispersion diagram, but will appear to repel each other as they start to get close on changing wave vector. The apparent near-crossing, called **anti-crossing**, of the LO and TO modes, and of the LA and TA modes, is also seen for the dispersion curves along [110].

8.5.8 The lattice dynamics of quartz

We choose the example of quartz to show the dispersion curves of a more complex insulating material. The bonds in quartz show elements of both covalent and ionic bonding. The SiO_4 tetrahedra are held together by strong covalent bonds, but the ions do not lose all their charge, so there are important electrostatic interactions. The dispersion curves have been measured by inelastic neutron scattering, and are shown in Fig. 8.17. Quartz is a good example because it is rather more complex than the alkali halides without being too complicated. There are nine atoms in the unit cell, so there are 27 dispersion curves (three acoustic, 24 optic).

Quartz has trigonal symmetry. For wave vectors along [001] there is a three-fold rotation axis, with three types of symmetry for the vibrations. In Fig. 8.17

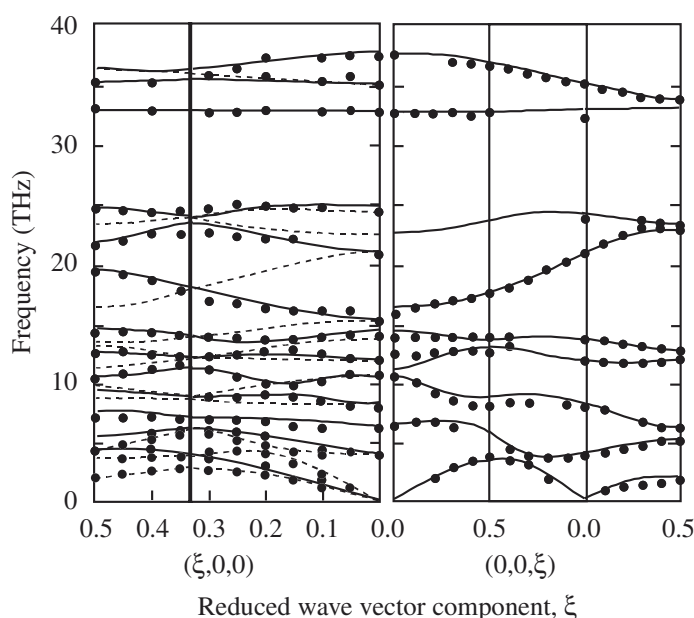


Fig. 8.17 Dispersion curves for quartz along two symmetry directions, noting the extended mode scheme for the [001] data, and the presence of two symmetry representations for the [100] data. The curves represent the results of a harmonic lattice dynamics calculation using force constants whose values have been fitted against the data. (Data are taken from Strauch and Dörner, *J. Phys.: Cond. Matter*, **5**, 6149, 1993.)

the modes of different symmetry for the [001] direction are plotted on separate diagrams, called an **extended zone scheme**. The order of these diagrams reflects the cases where modes become degenerate for the wave vectors exactly at the $k = 0$ or at the Brillouin zone boundary. For vibrations along [100] there are only two types of symmetry, and the dispersion curves are plotted on the same diagram.

The data for quartz show how complex the dispersion curves can be for complicated crystal structures. Some of the branches have quite a lot of variation of their frequency, whereas others are more constant. The frequency range extends nearly to 40 THz. This is five times larger than in NaCl, and since ω^2 is directly proportional to a force constant, it can be deduced that there are forces that are 25 times stronger in quartz than in NaCl. This is a reflection of the covalent bonds in quartz, which are particularly strong for vibrations involving stretching of the Si–O bonds.

The measurements of the dispersion curves in quartz represent a significant achievement. As will be shown in Chapter 10, neutron scattering methods are much easier for low-frequency modes, and often it is not possible to measure dispersion curves across the whole range of frequencies.

Summary of chapter

- The key assumption in the development of the theory of lattice dynamics is that atomic displacements are small so that the energy can be described as a **harmonic** function of the relative displacements of neighbouring atoms. Furthermore, it is assumed that if this approximation needs to be relaxed, it will be possible to modify the results of the harmonic model to account for the higher-order contributions to the energy.

- Within the **harmonic approximation**, equations of motion for all atoms are solved assuming that the atomic displacements can be described by **harmonic travelling waves**. The solutions of the equations give the frequency as a function of wave vector, $\omega(\mathbf{k})$. Plots of $\omega(\mathbf{k})$ are called **dispersion curves**. For systems with more than one atom in the unit cell, the solutions of the equations of motion will also give the relative displacement of each atom.
- Models of crystals with one atom in the unit cell give dispersion curves for which $\omega(\mathbf{k}) \rightarrow 0$ as $\mathbf{k} \rightarrow 0$. These solutions correspond to sound waves, and are called **acoustic modes**. Models with d dimensions have d acoustic modes.
- Models for three-dimensional monatomic crystals are solved and compared with experimental measurements of dispersion curves. The main features of the dispersion curves can be understood from the dominance of the **nearest-neighbour interatomic interactions**.
- Models with Z atoms and d dimensions have dZ modes of vibration for each wave vector, of which d are the acoustic modes. The acoustic modes are the vibrations for small $\mathbf{k}$ in which all the atoms in the unit cell move in phase, and the other modes represent all other combinations of the phase of atomic displacements. The vibrations in which atoms in the unit cell move with some atoms out of phase with others are called the **optic modes**.
- In general $\partial\omega^2/\partial\mathbf{k} = 0$ for $\mathbf{k} = 0$ and $\mathbf{k}$ half-way between reciprocal lattice vectors. The exception is where two modes become degenerate only at the latter point, in which case the sum of $\partial\omega/\partial\mathbf{k}$ for the two modes is equal to zero.
- The equations of motion can be conveniently represented by the formalism of the **dynamical matrix**, $\mathbf{D}(\mathbf{k})$. The eigenvalues of $\mathbf{D}(\mathbf{k})$ give $\omega^2(\mathbf{k})$, and the eigenvectors give the relative atomic displacements of each mode.

Further reading

Introductions of the subject of lattice dynamics are given by Kittel (1996) and Ashcroft and Mermin (1976). More detailed treatments of the topic are given by Barron and White (1999), Böttger (1983), Di Bartolo and Powell (1976), Dove (1993), Elliot (1998), Hardy and Karo (1979), Kosevich (1990), and Ziman (2000). Ferroelectric materials are discussed in detail by Lines and Glass (1977) and Mitsui and Tatsuzaki (1976).

Exercises

- (8.1) From a sketch of the atomic displacements associated with the $[110]$ and $[\bar{1}\bar{1}0]$ zone boundary acoustic modes in a ccp

crystal, demonstrate that the LA and one of the TA modes are equivalent at the $\langle 110 \rangle$ zone boundary wave vector.

- (8.2) Derive eqn 8.31 for the nearest-neighbour interaction. (*Hint*: start with $\partial r/\partial x = x/r$, which comes from $r^2 = x^2 + y^2 + z^2$, and note that for models with only nearest-neighbour interactions $\partial\phi/\partial r = 0$.)
- (8.3) Derive the expressions for the transverse and longitudinal interplanar force constants for neon for wave vectors along the (a) [100], (b) [111], and (c) [110] directions as given in eqns 8.32, 8.33–8.4.1. (*Hint*: for each direction of the wave vector use a set of axes in which the longitudinal direction is along one of the new axes, and express the vectors between an atom and its neighbours as a vector in this new set of axes.)
- (8.4) From the dispersion curves for neon (Fig. 8.7) obtain an estimate for the nearest-neighbour interatomic force constant K (Section 8.4.1). Assuming that the interaction between atoms can be represented using a Lennard-Jones potential 5.23, show that

$$K = \frac{72\epsilon}{2^{1/3}\sigma^2}$$

Compare your estimate of the value of K with that calculated from values of ϵ and σ obtained in Problem 5.1.

- (8.5) The model of the monatomic chain (Section 8.3) can be extended by including a local interaction for each atom of the form

$$E_{\text{local}}(u) = \frac{1}{2}\alpha u^2$$

Show that the effect of an additional local potential energy is to shift the dispersion curve for the case with nearest-neighbour interactions to

$$m\omega^2 = \alpha + 4J \sin^2 ka/2 \quad (8.62)$$

- (8.6) Explain why the LA(100) and TA(100) modes for potassium (Fig. 8.10) have very similar dispersion curves, by deriving the expressions for the interplanar force constants based on nearest-neighbour interatomic interactions given in Section 8.4.3.
- (8.7) Estimate the relative sizes of the three force constants that describe the dispersion curve of the LA mode along [111] in potassium (Fig. 8.10, Section 8.4.3). Explain this result.

- (8.8) The crystal structure of the linear molecule HCN is I -centred tetragonal, with one molecule per lattice point oriented along [001]. The two-dimensional **a**–**b** section of the crystal is shown below. If there is a single force constant between nearest-neighbour molecules for motions in the **a**–**b** plane as shown in the structure plan, show that (a) the frequencies of the LA and TA(y) modes along [100] are identical; (b) the frequency of the LA mode along [110] reaches the same maximum value as the LA model along [100]; (c) the TA(x – y) model for all wave vectors along [110] has zero frequency within the approximation of a single force constant. The latter result gives a ferroelastic phase transition in this material at a temperature of 170 K (Section 12.1.2).

- (8.9) Repeat the analysis leading to eqn 8.46 from the equation of motion for the small atom in the model for the diatomic chain (Section 8.5), showing that the same result is obtained.

- (8.10) From the general solution for the harmonic diatomic chain with unequal masses, eqn 8.4.9:

$$\omega^2 = \frac{J(M+m)}{Mn} \pm \frac{J}{Mn} \sqrt{(M+m)^2 - 4Mm \sin^2(ka/2)}$$

obtain the acoustic and optic mode solutions in the limit $k \rightarrow 0$ and at $k = \pi/a$.

- (8.11) Recast the harmonic model for the diatomic chain (Section 8.5) for the case where the atoms are equivalent, but with different force constants between the two neighbours of each atom, and obtain the solution analogous to eqn 8.49. Such a model could represent the lattice dynamics of the diamond or hcp structures. Show that the atomic displacements of the two modes at $k = \pi/a$ are different from the case of unequal masses but identical force constants (eqn 8.51).
- (8.12) Compare the measured dispersion curves of NaCl (Fig. 8.16) for a wave vector along [111] with the prediction of the simple diatomic chain (eqn 8.49).
- (8.13) Show that the one-dimensional dynamical matrix for the crystal of Fig. 8.15 has the form given by eqn 8.60.